Globalization techniques for MSPIN on phase-field fracture models

Conor McCoid

May 21st, 2026

Joint work with Blaise Bourdin at McMaster University

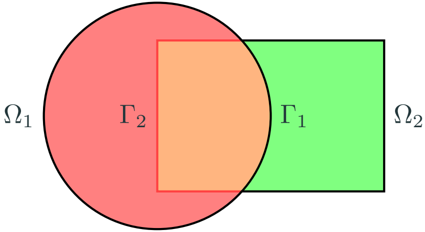
Outline

Introduction to domain decomposition and Schwarz methods

H.A. Schwarz, 1869

How do we solve the Laplace equation ($\Delta u = 0$) on complicated domains?

We split the domain into simpler subdomains.

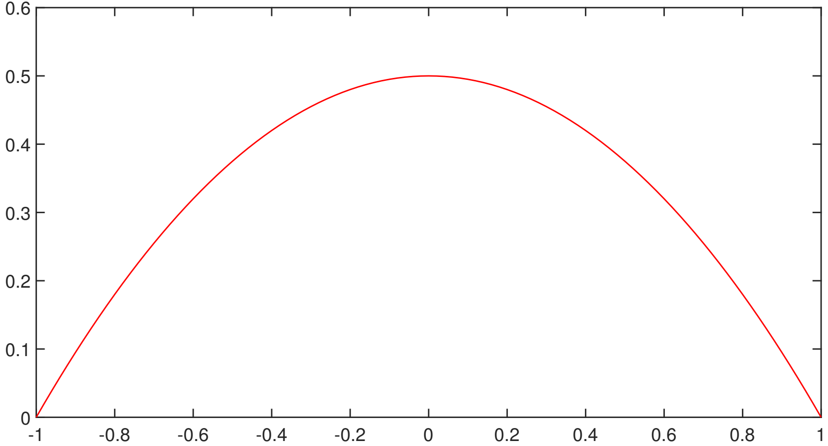


Alternating Schwarz method:

$$\begin{cases} \Delta u_1^{n+1} = 0 & \text{in } \Omega_1, \\ u_1^{n+1} = u_2^n & \text{on } \Gamma_1, \end{cases} \quad \begin{cases} \Delta u_2^{n+1} = 0 & \text{in } \Omega_2, \\ u_2^{n+1} = u_1^{n+1} & \text{on } \Gamma_2. \end{cases}$$

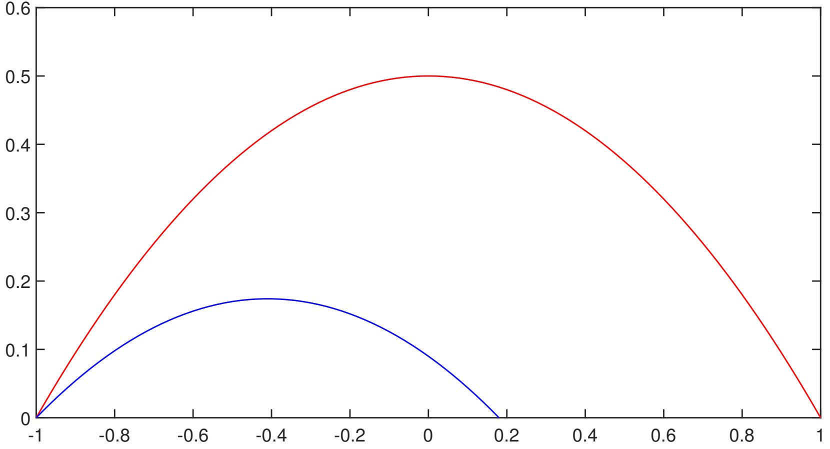
Simple example

$$u''(x) = -1, \quad x \in [-1,1], \quad u(-1) = u(1) = 0$$
$$\Omega_1 = [-1, 0.18], \quad \Omega_2 = [-0.22, 1]$$



Simple example

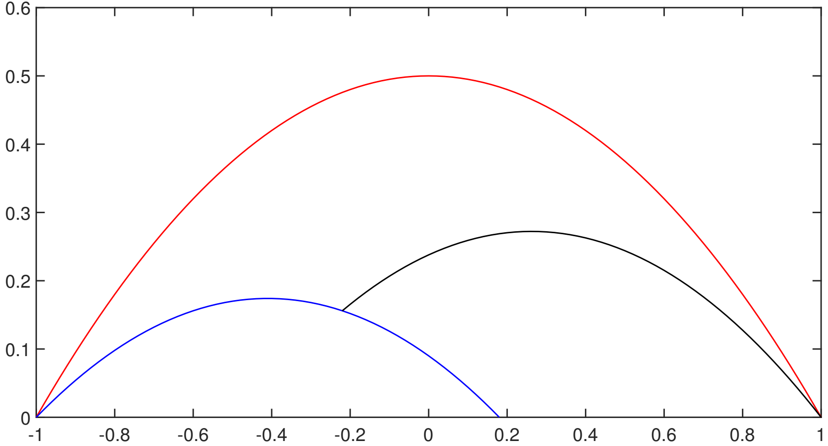
$$u''(x) = -1, \quad x \in [-1,1], \quad u(-1) = u(1) = 0$$
$$\Omega_1 = [-1, 0.18], \quad \Omega_2 = [-0.22, 1]$$



Simple example

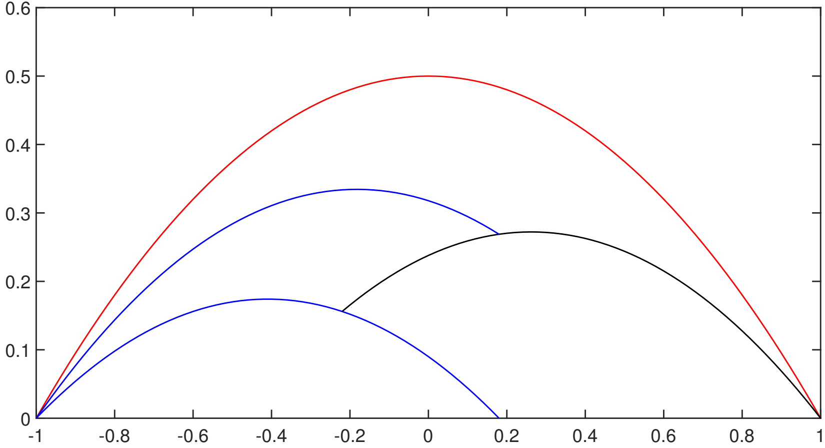
$$u''(x) = -1, \quad x \in [-1, 1], \quad u(-1) = u(1) = 0$$

$$\Omega_1 = [-1, 0.18], \quad \Omega_2 = [-0.22, 1]$$



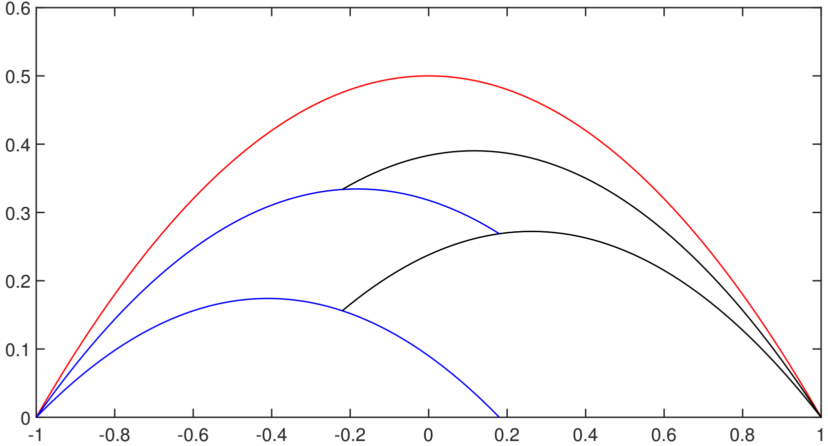
Simple example

$$u''(x) = -1, \quad x \in [-1, 1], \quad u(-1) = u(1) = 0$$
$$\Omega_1 = [-1, 0.18], \quad \Omega_2 = [-0.22, 1]$$



Simple example

$$u''(x) = -1, \quad x \in [-1, 1], \quad u(-1) = u(1) = 0$$
$$\Omega_1 = [-1, 0.18], \quad \Omega_2 = [-0.22, 1]$$



Field-split Schwarz methods

Instead of splitting up the physical domain, we can split up the problem into fields.

Suppose a problem can be written as

$$\begin{bmatrix} F(u,v) \\ G(u,v) \end{bmatrix} = \begin{bmatrix} f \\ g \end{bmatrix},$$

where F depends more strongly on u and G depends more strongly on v.

Field-split multiplicative Schwarz:

$$F(u^{(n+1)}, v^{(n)}) = f, \quad G(u^{(n+1)}, v^{(n+1)}) = g.$$

Phasefield fracture and Newton-Schwarz

Brittle fracture

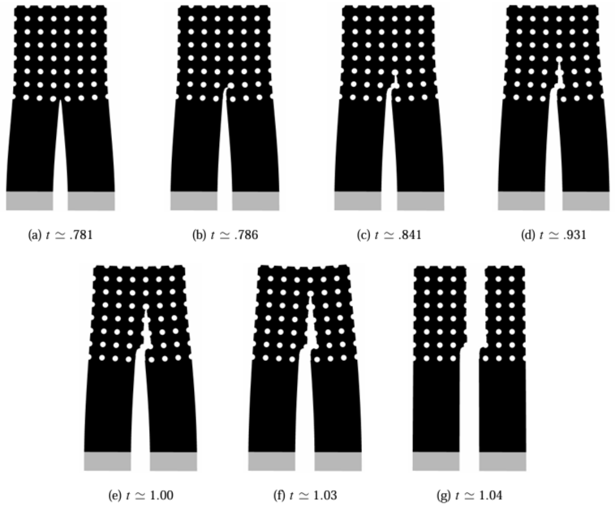


Figure 1: Crack propagation in brittle material, image taken from paper by Blaise Bourdin

Phasefield fracture model

$$E_\ell(u,\alpha) = \int \frac{1}{2}(1-\alpha)^2 Ae(u)\cdot e(u)dx + \frac{G_c}{4c_w}\int \frac{w(\alpha)}{\ell} + \ell\left|\nabla\alpha\right|^2 dx$$

- u is a vector field which defines the displacement
- α is a scalar field that is 0 away from the crack and 1 on the crack
- $e(u)$ is the strain tensor, A the stiffness tensor
- ℓ determines the accuracy of the approximation of the Hausdorff measure of the crack
- $w(0)=0$, $w(1)=1$, $w'(x)\geq 0$, $w\in C^1(0,1)$, $c_w=\int_0^1\sqrt{w(s)}ds$
- G_c is the critical energy release rate
- The model is quasi-static: energy is minimized for a given time step/external work, then the model steps forward in time

Alternate minimization

Cracks propagate to minimize $E_\ell(u,\alpha)$. To model this, we then need to minimize over both u and α:

$$\begin{bmatrix} F(u,v) \\ G(u,v) \end{bmatrix} = \begin{bmatrix} \frac{\partial}{\partial u} E_\ell(u,\alpha) \\ \frac{\partial}{\partial \alpha} E_\ell(u,\alpha) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

The energy E_ℓ is not convex in both variables, but it is convex if either u or α is kept constant. This leads to alternate minimization.

Algorithm 1 Alternate Minimization (AltMin)

1: Make initial guess α_0 and set $n = 0$
2: **while** $\|\alpha_{n+1} - \alpha_n\|$ is greater than tolerance **do**
3: Find $u_{n+1} = \operatorname{argmin}_u E_\ell(u, \alpha_n)$
4: Find $\alpha_{n+1} = \operatorname{argmin}_\alpha E_\ell(u_{n+1}, \alpha)$
5: **end while**

(animation)

Applying Newton (Kopanicakova, Kothari & Krause, 2023)

AltMin can be thought of as a fixed point iteration over α:

$$\alpha_{n+1} = \mathsf{AltMin}(\alpha_n),$$

which stops when α_{n+1} is sufficiently close to α_n.

To accelerate the otherwise linear convergence, we can apply Newton's method to AltMin(α) - α after the minimization steps:

$$\begin{bmatrix} J_{uu} & \\ J_{\alpha u} & J_{\alpha\alpha} \end{bmatrix}^{-1} \begin{bmatrix} J_{uu} & J_{u\alpha} \\ J_{\alpha u} & J_{\alpha\alpha} \end{bmatrix} \begin{bmatrix} u_* - u_n \\ \alpha_* - \alpha_n \end{bmatrix} = \begin{bmatrix} u_{n+1} - u_n \\ \alpha_{n+1} - \alpha_n \end{bmatrix},$$

where J_{ij} is the second derivative of $E_\ell(u, \alpha)$ with respect to i then j.

In practice, we solve this system inexactly and usually with some kind of backtracking to globalize Newton's method. The resulting method is called multiplicative Schwarz preconditioning inexact Newton's method (MSPIN).

Globalization techniques

Newton's method needs globalization techniques to converge when the starting guess is far from a minimum. The default choice is cubic backtracking:

Algorithm 2 Cubic backtracking

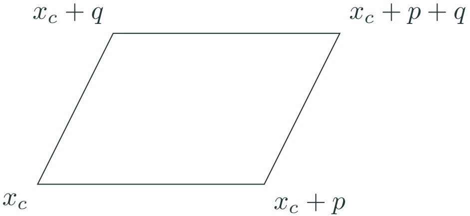
Inputs: current iterate x_c, search direction p, objective function $F(x)$
Set $\lambda_0 = 1$
Set λ_1 such that $x_p = x_c + \lambda_1 p$ minimizes the quadratic polynomial interpolating $F(x_c)$, $F(x_c + p)$ and $\nabla F(x_c)^\top p$
while $F(x_c + \lambda_1 p) \geq F(x_c)$ and $\|p\| <$ some tolerance **do**
    Set λ_2 such that $x_c + \lambda_2 p$ minimizes the cubic polynomial interpolating $F(x_c)$, $F(x_c + \lambda_0 p)$, $F(x_c + \lambda_1 p)$ and $\nabla F(x_c)^\top p$
    $\lambda_0 \leftarrow \lambda_1$, $\lambda_1 \leftarrow \lambda_2$
end while

Parallelogram minimization

MSPIN is well-suited for a new kind of globalization technique because it requires finding two solutions for the same problem:

- the result from multiplicative Schwarz/AltMin, $x_c + q$, and;
- the result from the Newton's method step, $x_c + p$.

This gives us two directions to look in, meaning we should seek our minimum in a parallelogram.



Parallelogram minimization

Given an objective function $F(x)$, minimize the polynomial $P(i,j) = ai^2 + bij + cj^2 + di + ej + f$, where

$$\begin{aligned} f =& F(x_c), & e =& \nabla F(x_c)^\top p, & d =& \nabla F(x_c)^\top q, \\ c =& F(x_c + p) - e - f, & a =& F(x_c + q) - c - f, \end{aligned}$$

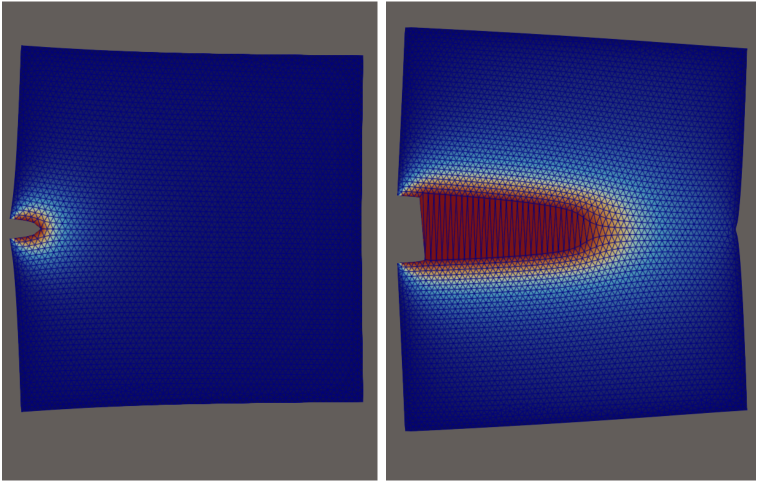
$$b = F(x_c + p + q) - F(x_c + p) - F(x_c + q) + f,$$

in the region where $0 < i, j < 1$.

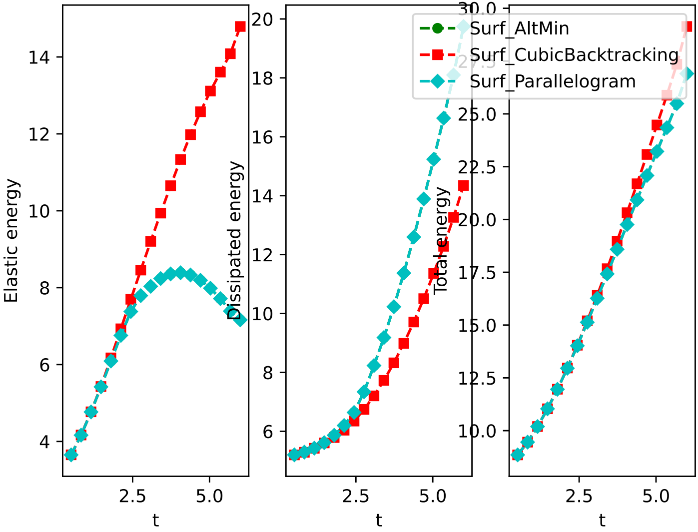
$$\begin{bmatrix} i_{\mathsf{min}} \\ j_{\mathsf{min}} \end{bmatrix} \in \left\{ \begin{bmatrix} \frac{-2cd+be}{4ac-b^2} \\ \frac{-2ae+bd}{4ac-b^2} \end{bmatrix}, \begin{bmatrix} 0 \\ -\frac{e}{2c} \end{bmatrix}, \begin{bmatrix} -\frac{d}{2a} \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ -\frac{e+b}{2c} \end{bmatrix}, \begin{bmatrix} -\frac{d+b}{2a} \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\}$$

Our new minimum is $x_n = x_c + i_{\mathsf{min}} q + j_{\mathsf{min}} p$.

Example calculations

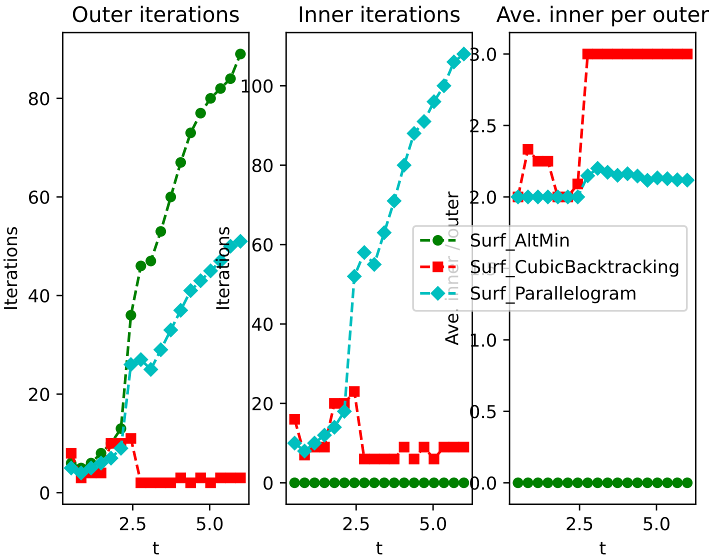


Energy



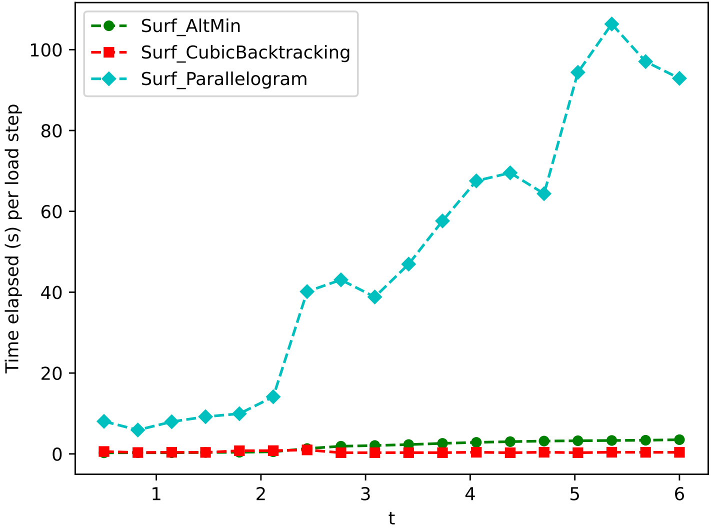
Elastic energy
Dissipated energy
Total energy
t
t
t
Surf_AltMin
Surf_CubicBacktracking
Surf_Parallelogram

Iterations



Outer iterations
Inner iterations
Ave. inner per outer
Iterations
Ave. inner / outer
t
0
20
40
60
80
100
0.0
0.5
1.0
1.5
2.0
2.5
3.0
2.5
5.0
Surf_AltMin
Surf_CubicBacktracking
Surf_Parallelogram

Time


Surf_AltMin
Surf_CubicBacktracking
Surf_Parallelogram
Time elapsed (s) per load step
0
20
40
60
80
100
1
2
3
4
5
6
t

Summary

Domain decomposition and Schwarz methods permit parallelization and can improve problem conditioning. We need additional methods to accelerate convergence rates above linear.

Newton-Schwarz methods need globalization techniques to stabilize them for difficult problems. Parallelogram minimization combines both the Newton and Schwarz directions for faster convergence.

Thank you

References

- Bourdin, B. **Numerical implementation of the variational formulation for quasi-static brittle fracture.** *Interfaces and Free Boundaries* 9 (2007).
- Kopanicakova, A., Kothari, H. & Krause, R. **Nonlinear field-split preconditioners for solving monolithic phase-field models of brittle fracture.** *Computer methods in applied mechanics and engineering* 403 (2023).
- Cai, X.C. & Keyes, D.E. **Nonlinearly preconditioned inexact Newton algorithms.** *SIAM Journal on Scientific Computing* 24 (2002).